Idempotence

An operation is idempotent if performing it several times has the same effect as performing it once. Setting a key to a fixed value is idempotent; incrementing a counter is not.

Idempotence matters because networks are unreliable. When a client sends a request and the response times out, the client cannot know whether the server processed it. Retrying is only safe if the operation is idempotent.

Unique request IDs

A common technique is to attach a unique request ID to every operation. The server records the IDs it has already processed and ignores duplicates. Combined with retries, this gives effectively-once processing.

Summary

Make retried operations idempotent, or deduplicate them with request IDs, so that a timeout never causes a duplicate payment.